L1 SOC ALERT TRIAGE REPORT

Alert Report 04 – Linux User Account Creation and Group Modification Investigation

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1. ALERT SUMMARY

Alert Title:	Linux User Account Creation and Privileged Group Modification
Target Host:	`ubuntu-soc`
Log Source & Index:	`/var/log/auth.log` | Index: `main` (Sourcetype: `linux_secure`)
Created User Account:	`splunk` (UID: 1001 | GID: 1001)
Assigned Severity:	**LOW / INFORMATIONAL**
Triage Assessment:	**CLOSED - BENIGN ADMINISTRATIVE ACTIVITY**

2. ALERT DETAILS

The SIEM monitoring environment flagged a local user account creation event on `ubuntu-soc`. Account creation and group modification are security-sensitive actions because adversaries often create persistent user accounts or add users to privileged groups to maintain access. As an L1 analyst, I investigated the event details in Splunk Enterprise to verify whether this activity was benign or unauthorized.

3. INVESTIGATION OBJECTIVE

The objective of this investigation was to:

> Isolate the exact user creation command log in Splunk and extract account properties (Username, UID, GID, Home Directory, Shell). Identify subsequent group modifications and service operations associated with the created account. Correlate timestamps and process events to confirm whether the activity stems from authorized lab setup or unauthorized intrusion.

4. TOOLS USED

Component	Technical Details
Operating System	Ubuntu Linux running inside Oracle VirtualBox VM
SIEM Platform	Splunk Enterprise 10.6.1 (Access via `http://localhost:8000`)
Log File Monitored	`/var/log/auth.log` indexed under `main` / `linux_secure`

5. SPL QUERIES USED

`index=main sourcetype=linux_secure ("useradd" OR "adduser")` — Isolates user account creation log entries. `index=main sourcetype=linux_secure ("useradd" OR "adduser") | table _time host _raw | sort -_time` — Formats account creation events into a structured timeline view. `index=main sourcetype=linux_secure "splunk" | table _time host _raw | sort -_time` — Queries all 47 events mentioning the `splunk` account. — Filters 19 commands executed for or by the user. — Examines surrounding user sessions and failure logs (43 events).

```
index=main sourcetype=linux_secure "splunk" "COMMAND=" | table _time host _raw
| sort -_timesplunkindex=main sourcetype=linux_secure "gunjan" ("authentication
failure" OR "session opened" OR "session closed") | table _time host _raw |
sort -_time
```

6. EVIDENCE AND SCREENSHOT ANALYSIS

Step 1: Identifying User Account Creation

I searched Splunk for user creation entries using `("useradd" OR "adduser")`. Splunk returned **1 matching event** recorded at 12:40:11.545 IST on 5 November 2026.

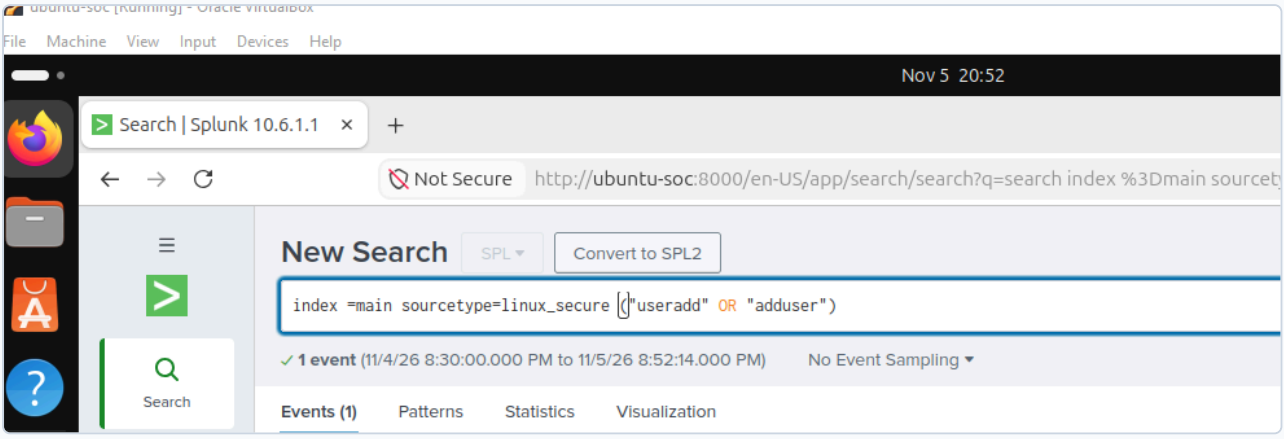


Figure 1: Splunk search returning 1 event for `useradd` log entry in `/var/log/auth.log`.
Shows raw event payload from `useradd[6140]` capturing new user parameters on host `ubuntu-soc`.

Step 2: Account Property Extraction

Formatting the result with `table _time host _raw` allowed exact extraction of user account parameters:

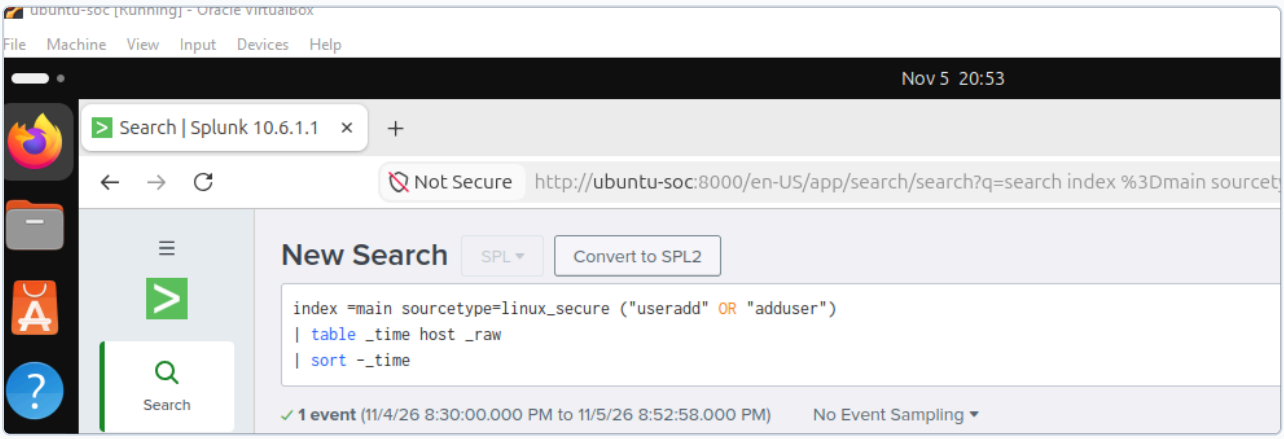


Figure 2: Formatted table view displaying the exact raw creation string and timestamp.
Raw Log: `useradd[6140]: new user: name=splunk, UID=1001, GID=1001, home=/opt/splunk, shell=/bin/bash, from=/dev/pts/1`.

Extracted Account Parameters:

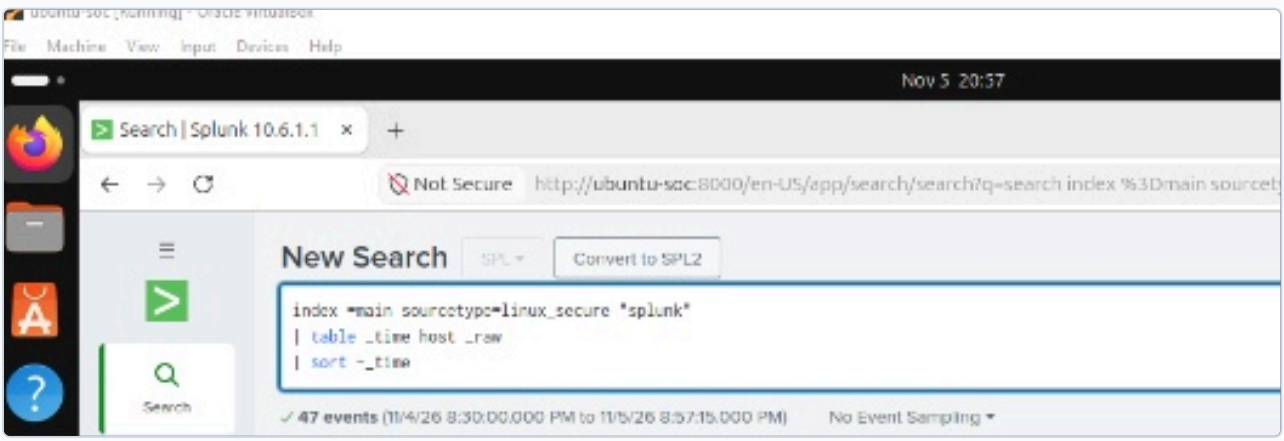


Figure 3: Search results returning 47 total events associated with the `splunk` account.

Displays session openings, service executions, and log closing entries for user `splunk`.

Step 4: Isolating Executed Elevated Commands

Adding the filter `"COMMAND="` returned **19 specific command execution events** associated with the `splunk` user.

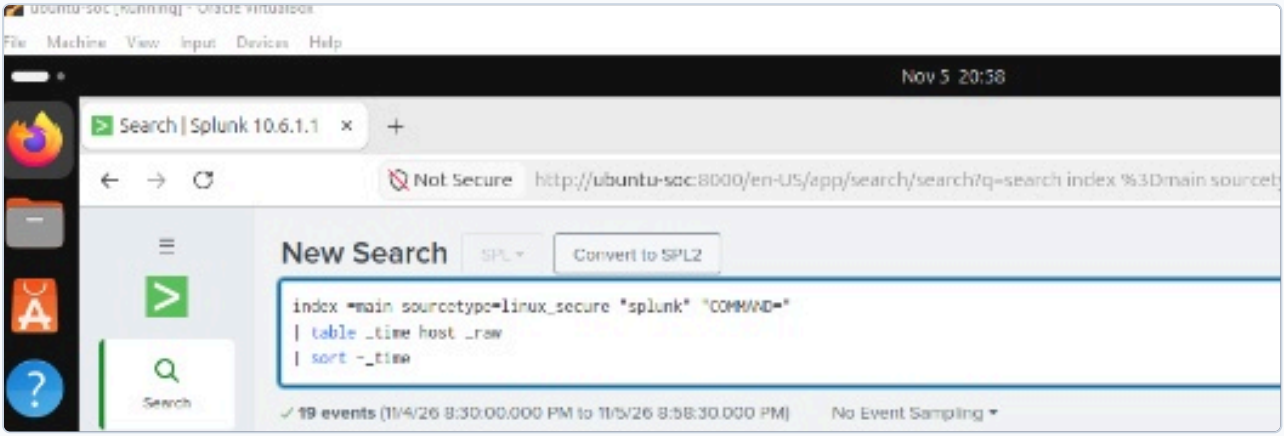


Figure 4: Formatted table displaying 19 command execution events involving the `splunk` account.

Shows repeated administrative service operations (`splunk start`, `splunk status`, `splunk stop`, and `usermod -aG adm splunk`).

Step 5: Reviewing Surrounding Sessions and Local Failures

I searched for session events and local authentication checks for user `gunjan` (43 matching events), confirming local interactive desktop usage without any remote connection strings.

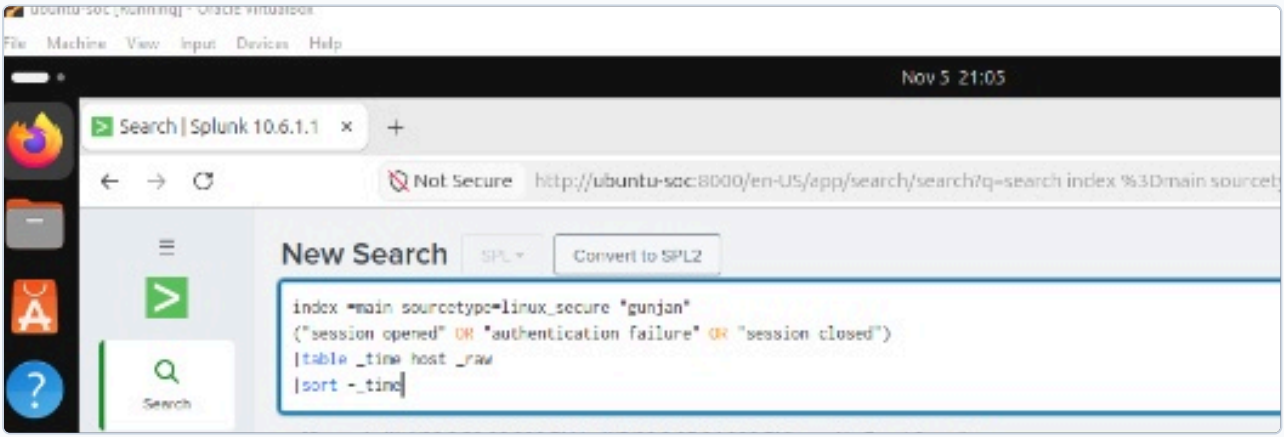


Figure 5: Session opened/closed logs and local PAM authentication checks for user `gunjan` (43 events).

Shows standard local desktop session events and local PAM password checks during admin operations.